

Adaptive Auto-Harness: Sustained Self-Improvement for Agentic System Deployment on Open-Ended Task Streams

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Abstract

Auto-harness systems such as A-Evolve, GEPA, and Meta-Harness improve LLM agents by optimizing prompts, skills, tools, memories, and supporting infrastructure from execution feedback, but they are typically evaluated on fixed offline benchmarks. Real deployments instead present open-ended task streams: histories grow without a fixed endpoint, heterogeneous tasks require different harnesses, and problem distributions shift over time. These challenges make a single repeatedly and densely updated harness brittle, causing performance degradation as accuracy peaks early and then declines. This motivates sustained harness construction with task-wise adaptation. We introduce Adaptive Auto-Harness, a framework and system for such streams. The framework decomposes the gap to an oracle harness into evolution loss and adaptation loss. The system addresses these losses with a stateful multi-agent evolver, a harness tree with solve-time routing, and human-steering hooks for cases where history lacks the needed signal. Across prediction-market, security-competition, and event-forecasting streams, Adaptive Auto-Harness outperforms five existing auto-harness baselines and ablations attribute gains to better construction, routing, or targeted human steering.

1 Introduction

Open-ended task streams are a common deployment regime for LLM agents: tasks arrive continuously, feedback accumulates over time, and future tasks may differ from earlier ones. In this regime, an agent’s harness, comprising the prompts, skills, tools, and supporting infrastructure that surround a fixed LLM, is a primary determinant of task-solving performance. *Auto-harness systems* such as A-Evolve (Lin et al., 2026), GEPA (Agrawal et al., 2025), and Meta-Harness (Lee et al., 2026) perform evolution to construct harness automatically

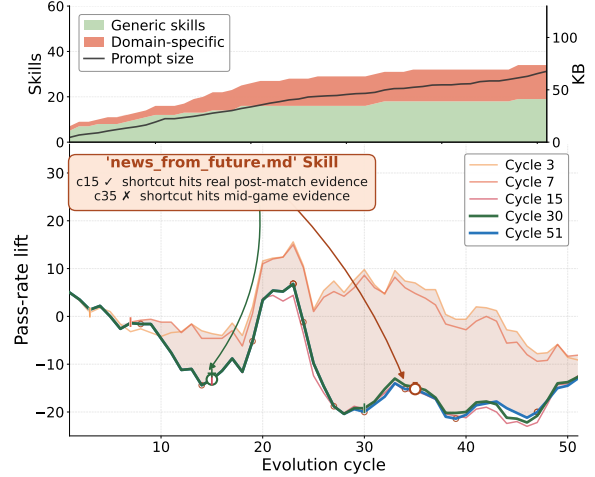


Figure 1: Longer one-shot evolution can overfit earlier stream evidence. **Top:** The unbounded A-Evolve run grows from 12 to 34 skills, while the prompt grows from 2 KB to 68 KB. **Bottom:** Each curve reports pass-rate increase relative to the no-evolution solver, with different evolution stopping cycles. Early gains fade as later tasks arrive; `news_from_future.md` marks an overfitting example of the evolved skill.

from execution feedback, and report substantial gains on static offline benchmarks such as SWE-bench (Jimenez et al., 2024). Those evaluations, however, do not capture the central pressure of deployment: the harness must keep improving while operating on a chronological stream whose history grows, task types vary, and distribution shifts.

Representative streams include prediction markets with thousands of questions over weeks (Cheng et al., 2026), decade-long security competitions (Zhuo et al., 2025), and cross-lingual forecasting services with heterogeneous sources (Zeng et al., 2025). Figure 1 illustrates why repeatedly enlarging one harness is insufficient. On Polymarket: we run A-Evolve and stop evolution after 3, 7, 15, 30, or 51 cycles, comparing each run with the same solver without evolution. Early evolution improves pass rate, but longer runs accumulate larger prompts and more specialized

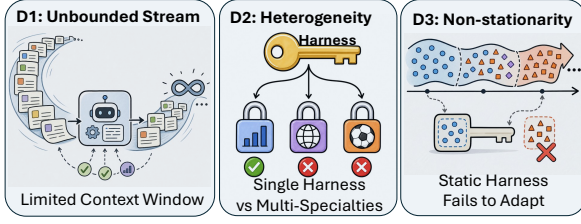


Figure 2: **Three deployment dimensions in open-ended task streams.** Unbounded stream, heterogeneous tasks, and non-stationary distributions expose the limits of evolving a single dense harness for long-term deployment

skills. For example, `news_from_future.md` first detects post-match evidence and later fires on mid-game evidence, turning a useful temporal cue into a stale shortcut. All stopping budgets eventually peak and decline; later in the stream, shorter runs outperform longer ones. Sustained deployment therefore requires preserving useful history while adapting the active harness to the current task.

This failure exposes three deployment dimensions that static benchmark evaluation does not capture (Figure 2). **(D1) Unbounded Streams.** The task stream has no fixed train/test cutoff or designated endpoint (Karten et al., 2026; Wang et al., 2023); feedback, trajectories, and harness state accumulate throughout deployment, creating a heavy burden for the evolution. Existing auto-harness systems built around a single-agent evolver compress this expanding history into a finite context window, creating a bottleneck for building the effective and generalizable harness. **(D2) Task heterogeneity.** Varied types of tasks are mixed in the same stream. A prediction-market platform, for example, mixes politics, sports, and finance questions in the same hour, each calling for distinct sources, tools, and prompting. However, existing auto-harness systems deploy a static dense harness across the stream, with no solve-time adaptation to the task at hand, and neglects the fact that a single fixed policy is rarely optimal across heterogeneous problems (Miao et al., 2025). **(D3) Distributional non-stationarity.** As the stream progresses, incoming tasks shift away from the experience the harness was last fitted on. A harness optimized for recent cycles therefore drifts out of fit for new tasks, even with rich historical experience. Closing this gap requires per-task contextual adaptation of the harness, not only continued historical fitting. Additional diagnostics are shown in Appendix A.

We address these dimensions with a unified analytical framework and identify two root gaps that prior auto-harness systems overlook. This analysis motivates: (1) *Sustained auto-harness* (§3.3), which replaces stateless one-shot evolution with a stateful multi-agent system with cross-cycle knowledge for better harness construction; and (2) *Solve-time adaptation* (§3.4), which adapts a task-relevant harness for each problem prior to solving, restoring per-task fit on a heterogeneous, drifting stream. Beyond the two axes, we further introduce a third axis: human-in-the-loop (HITL) for auxiliary steering of the harness to incorporate human insights and provisions that are absent from historical experience. Our contributions are:

- (1) **Deployment-regime analysis of open-ended agentic streams.** We formalise why static auto-harness evaluation is insufficient once tasks arrive as unbounded, heterogeneous, and non-stationary streams. The framework decomposes the gap towards the optimal harness into evolution loss and adaptation loss, providing guidance for auto-harness system designs.
- (2) **Adaptive Auto-harness system.** We introduce a stateful multi-agent evolver for sustained harness construction, a harness-tree router for solve-time adaptation, and structurally triggered human-in-the-loop hooks for regimes where historical experience is insufficient.
- (3) **Comprehensive empirical validation and diagnosis.** We evaluate on three streaming tasks spanning prediction markets, security challenges, and event forecasting against other auto-harness systems. Beyond aggregated performance, we provide in-depth analysis and evidence for the proposed gaps, component ablations, and human-in-the-loop slice analyses.

2 Related Work

Continual Learning in Task Streams. Continual learning studies systems that learn from a sequence of tasks while retaining earlier capabilities (Buzzega et al., 2020; Wang et al., 2022b,a). Domain and test-time adaptation address distribution shift (Ben-David et al., 2010; Ganin et al., 2016; Wang et al., 2021; Liang et al., 2020), and mixture-of-experts methods route heterogeneous inputs to specialised components (Jacobs et al., 1991; Shazeer et al., 2017). These directions cover pieces of our D1–D3 setting, but they usually adapt model weights, classifiers, or expert modules. In

contrast, we study the harness-level analogue.

Self-improving and self-evolving agents. The closest LLM-agent precedents to our setting are systems that update the harness directly from execution feedback. A-Evolve (Lin et al., 2026) introduces a linear-chain evolver: each cycle reads batch trajectories and mutates prompts, skills, memory, and tools. GEPA (Agrawal et al., 2025) adds reflective Pareto prompt evolution using textual feedback rather than scalar reward. Meta-Harness (Lee et al., 2026) uses a growing filesystem archive and Claude Code as the proposer. Continual Harness (Karten et al., 2026) enables online adaptation within a single continuous deployment run through alternating action/refinement cycles. SkillOS (Ouyang et al., 2026) learns a skill curation policy via reinforcement learning, training a curator to select and refine reusable skills from repeated interactions.

3 Method

3.1 Problem Formulation

We consider tasks arriving in an open-ended stream $x_1, x_2, \dots, x_T$ with $x_t \sim P_t$, where P_t is the task distribution at time t and each x_t has a fixed ground truth $y(x_t)$. The agent observes the history of prior experience $\mathcal{H}_t = \{(x_i, r_i, \tau_i)\}_{i=1}^{t-1}$, where r_i is optionally the realized reward on task x_i and τ_i is the agent’s solving trajectory (actions, intermediate observations, tool calls). A *harness* $C = \varphi(\mathcal{H}_t)$, with $|C| \leq K$, is a bounded representation (prompts, skills, memory, and tools) with capacity budget K , where φ is the *evolver*: the agentic system that automatically transforms historic experience into the harness. A solver agent then acts via the policy $a_t \sim \pi(a \mid x_t, C_t)$, where π is the LLM-induced sampling distribution over actions conditioned on the task and harness.

3.2 Analytical Framing

The three deployment dimensions of §1 surface concrete failures, but they do not by themselves identify what an evolver should fix. To pinpoint the root causes, we frame the problem analytically: we cast harness construction as a regret-minimization problem against an oracle reference, and decompose the regret into two complementary loss terms that map directly to actionable design axes.

Utility and regret. We define the utility of a harness C on a task x_t as

$$V(C, x_t) = \mathbb{E}_{a \sim \pi(\cdot \mid x_t, C)}[r(a, y(x_t))], \quad (1)$$

which is the expected reward of the harness-conditioned solver. The full-history utility is the corresponding ceiling under the capacity budget, $V(\mathcal{H}_t, x_t) := \sup_{C: |C| \leq K} V(C, x_t)$, that is, the best utility attainable by any bounded harness on x_t . The regret of the evolver-constructed harness is then

$$\text{Regret}(\varphi, x_t) = V(\mathcal{H}_t, x_t) - V(\varphi(\mathcal{H}_t), x_t) \geq 0, \quad (2)$$

non-negative since $\varphi(\mathcal{H}_t)$ is one element of the supremum’s domain. Operationally, this ceiling matches what a solver granted $\mathcal{H}_t$ directly, under the same compute and tool budget as φ , could attain by reconstructing any candidate harness on the fly. **Solve-time optimal harness.** Fix the evolver class Φ and define the solve-time optimal harness for a particular task:

$$C_\Phi^*(x_t) = \arg \max_{\substack{C = \varphi(\mathcal{H}_t, x_t), \varphi \in \Phi \\ |C| \leq K}} V(C, x_t). \quad (3)$$

This is an hypothetical oracle reference: it is the best harness the evolver class Φ could produce if it were allowed to condition on the incoming task x_t at solve time. A deployed evolver φ commits to one harness $\varphi(\mathcal{H}_t)$ before x_t is observed; using C_Φ^* as pivot between $V(\mathcal{H}_t, x_t)$ and $V(\varphi(\mathcal{H}_t), x_t)$ yields the following decomposition.

Proposition 1 (Regret Decomposition). *For any deployed evolver $\varphi \in \Phi$,*

$$\mathbb{E}_{x_t}[\text{Regret}(\varphi, x_t)] = L_{\text{evo}}(\Phi) + L_{\text{adapt}}(\varphi), \quad (4)$$

where

$$L_{\text{evo}}(\Phi) = \mathbb{E}_{x_t}[V(\mathcal{H}_t, x_t) - V(C_\Phi^*(x_t), x_t)], \quad (5)$$

$$L_{\text{adapt}}(\varphi) = \mathbb{E}_{x_t}[V(C_\Phi^*(x_t), x_t) - V(\varphi(\mathcal{H}_t), x_t)]. \quad (6)$$

L_{evo} is the *evolution loss*: it reflects the evolver class Φ ’s capability gap, regardless of how many cycles the evolver runs. A single-agent prompt editor cannot produce multi-file infrastructure; that ceiling is structural, not a matter of effort. Reducing L_{evo} therefore requires pursuing more capable evolver systems with access to broader control and feedbacks. L_{adapt} is the *adaptation loss*: the oracle builds the optimal harness per task, but a deployed φ commits to one harness before seeing x_t . Therefore, even with an optimal evolver system, L_{adapt} exists as long as task heterogeneity persists.

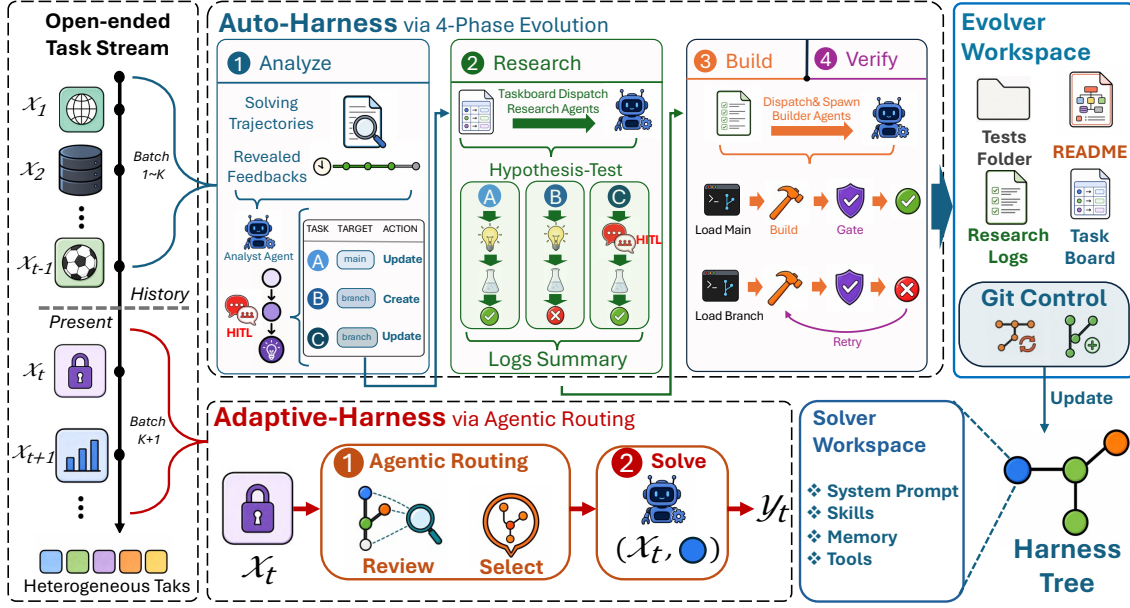


Figure 3: Overview of the Adaptive Auto-Harness system. **Bottom:** the multi-agent evolver constructs and refines a harness tree across cycles via four roles (Analyst $\rightarrow$ parallel Researchers $\rightarrow$ Builder $\rightarrow$ Verifier) with a persistent cross-cycle workspace and temporal-reveal feedback. **Top:** at solve time, a router agent reads each branch’s workspace via git show and routes the incoming task x_t to the most suitable branch. Two human-in-the-loop hooks (task-board steering and research-phase assistance) trigger only when the evolver’s history lacks relevant signal.

Human-in-the-loop as a third axis. The decomposition assumes $\mathcal{H}_t$ contains relevant signal for x_t . When it does not, the regret decomposition no longer applies; we address this case via a third axis outside $L_{\text{evo}} + L_{\text{adapt}}$, the human-in-the-loop channel (§3.5).

3.3 Sustained Auto-Harness via Multi-Agent Evolution

The analytical decomposition identifies L_{evo} as the loss from harness capabilities that the evolver class cannot construct from history. We reduce this loss by expanding Φ with a stateful four-phase multi-agent evolver, temporal-reveal feedback, and cross-cycle memory (Figure 3). This design targets the unbounded-stream failure where a single-agent evolver must absorb growing trajectories, delayed labels, and prior research within one context window. Concretely, we address three structural limitations:

(1) Multi-agent with distinct roles and objectives. Because unbounded streams (D1) keep expanding the trajectory history an evolver must interpret, existing single-agent evolvers must fit analysis, research, implementation, and verification into one context window. We decompose evolution into four agents (Analyst $\rightarrow$ parallel Researchers $\rightarrow$ Builder $\rightarrow$ Verifier), each with a dedicated objective and

full context budget. This eliminates the single-window bottleneck and lets parallel Researchers explore disjoint hypotheses without premature convergence.

(2) Temporal-reveal feedback. Under unbounded streams (D1), labels arrive asynchronously (e.g., a prediction market resolves days after the trade). We implement a *temporal-reveal gate* that surfaces each task’s ground-truth label to the evolver only after its resolution date, providing a proper streaming feedback signal without leaking future information.

(3) Persistent cross-cycle state. We provide the evolver with a dedicated workspace that persists across cycles, containing a task board (prioritised failure analysis), research logs (tested hypotheses with pass/fail verdicts), architecture documentation, and verification tests. This cross-cycle memory enables the evolver to refine its construction ability over time and build upon prior evolution experience rather than restarting from scratch.

3.4 Solve-Time Adaptation via Harness-Tree Routing

The analytical decomposition identifies L_{adapt} as the loss from committing to one dense harness before observing the incoming task’s context. We reduce this loss by shifting heavy adaptation into evolution time: the evolver constructs a *harness*

tree of specialized harnesses, and solve time only requires a lightweight routing decision (Figure 3, top panel). This design targets task heterogeneity and non-stationarity, where a single dense harness can underfit specialized task contexts even when the evolver is capable.

We instantiate this with two designs (Figure 3, top panel). **(1) Branching harness tree (built at evolution time).** The solver’s workspace is a git repository. The evolver constructs regime-specific branches (e.g., branch/crypto-classical, branch/binary-reversing) during evolution, each carrying its own prompt, skills, and tool registry; git provides versioning, isolation, and lineage tracking across branches. **(2) Agentic routing (executed at solve time).** A router agent reads each branch’s workspace via `git show` and selects the branch given x_t ’s context; the solver then checks out that branch and executes.

3.5 Human-in-the-Loop Channel

A third failure lies outside $L_{\text{evo}} + L_{\text{adapt}}$: some tasks require harness absent from $\mathcal{H}_t$ signal, such as API credentials, novel web sources, or proprietary endpoints. In this experience-insufficient setting, neither a stronger evolver nor solve-time routing can recover the missing signal. We address it with a human-in-the-loop channel that augments $\mathcal{H}_t$ through structurally triggered steering hooks (Figure 3). This design targets open-ended streams where new access requirements appear before autonomous evolution has relevant evidence. **(1) Task-board steering.** After the Analyst updates the task board, a human may review it to add entries, adjust priorities, or supply domain guidance and source access. This proactively steers the subsequent research cycle with direction the evolver cannot derive from trajectories alone. **(2) Interactive assistance during research.** When a Researcher hits a barrier mid-execution that requires human intervention (e.g., an authentication wall), the hook prompts the human in real time. This reactively unblocks the research agent at the point of failure.

4 Experiments

4.1 Experimental Setup

Benchmarks. We evaluate on three open-ended task streams (Table 1): *PolyBench* for prediction markets (Cheng et al., 2026), *CTF-Dojo* for security challenges (Zhuo et al., 2025), and *FutureX*

Table 1: Benchmark statistics for the three chronological task streams.

Bench.	Tasks	Span	Resolution	Domain
PolyBench	5,075	Feb 6–22, 2026	Day	Prediction markets
CTF-Dojo	261	2011–2024	Year	Security
FutureX	503	Jan–Apr 2026	Week	Forecasting

for event forecasting (Zeng et al., 2025). All three enforce strict temporal order and covers the three dimensions of challenge. Details and non-stationarity diagnostics are in Appendix A and C.

Baselines. We compare with no-evolution runs with different solver agents using **Sonnet-4.6** (Anthropic, 2025b), **DeepSeek-V3.2** (DeepSeek-AI, 2025), **Claude Haiku** (Anthropic, 2025a), **textbfGLM-4.7** (Z.ai, 2025), and **Kimi-K2.5** (Moonshot AI, 2026). We compare with five auto-harness baselines: **A-Evolve** (Lin et al., 2026), **GEPA** (Agrawal et al., 2025), **Meta-Harness** (Lee et al., 2026), **Continual Harness**, and **SkillOS**. We also compare with one human-designed system **OctoTools** (Lu et al., 2025).

Solver and evolver. We use Claude Sonnet 4.6 as the solver for all experiments unless specified, and use Claude Opus 4.6 as the evolver, both at temperature 0 to attribute gains to the evolution algorithm rather than sampling noise; the no-evolution controls additionally report base-agent results for Haiku 4.5, DeepSeek-V3.2, Kimi-K2.5, and GLM-4.7. All algorithms share the same batch size (100/20/20 for PolyBench/CTF-Dojo/FutureX), batch loop, and temporal-reveal gate; only the evolution algorithm varies.

Metrics. For PolyBench, we report three complementary metrics: *Coverage*, the fraction of all 5,075 markets the agent traded; *Accuracy*, the fraction of all markets traded correctly; and *Return*, defined as $\text{Coverage} \times \text{CWR}$, where CWR is the confidence-weighted portfolio profit-to-investment ratio over traded markets. For CTF-Dojo and FutureX, we report the official pass rate (Pass@1) defined by the original benchmarks (Zhuo et al., 2025; Zeng et al., 2025). We also report *lift* in the figures as the difference with baselines. Full definitions are in Appendix A.

Research questions. **RQ1** (§4.2): How does Adaptive Auto-Harness compare against existing auto-harness systems on open-ended task streams? **RQ2** (§4.3): How does Adaptive Auto-Harness address benchmark-specific bottlenecks? **RQ3** (§4.4): Does stateful multi-agent evolution with evaluation

Table 2: Main comparison across no-evolution agents, auto-harness baselines, a human-designed system, and our three variants. PolyBench reports Coverage / Accuracy / Return (coverage-scaled CWR, in %); CTF-Dojo and FutureX report the official Pass@1. Bold marks the best result per row and underline marks second best.

Benchmark	Metric	No evolution					Auto-harness baselines					Human	Ours		
		Sonnet	Haiku	DeepSeek	Kimi	GLM	A-Evolve	GEPA	Meta Harness	Cont. Harness	SkilIOS	OctoTools	Multi agent	Adaptive	Full System
PolyBench	Coverage $\uparrow$	31.5	32.6	2.9	24.7	21.8	21.1	22.6	55.3	10.2	24.1	54.4	<u>95.8</u>	91.4	97.9
	Accuracy $\uparrow$	22.2	15.2	1.4	14.0	14.0	18.4	13.4	50.8	8.5	21.4	40.0	<u>79.8</u>	77.4	80.9
	Return $\uparrow$	+1.7	+88.0	+16.4	-2.0	+10.1	+7.2	+0.2	+320	+1.7	+3.6	+20.4	<u>+351</u>	+352	+330
CTF-Dojo	Pass $\uparrow$	37.2	23.8	26.1	24.5	12.6	45.2	42.9	41.0	25.7	29.5	38.3	<u>47.9</u>	46.0	50.2
FutureX	Pass $\uparrow$	31.0	31.0	31.2	27.8	30.8	<u>47.5</u>	28.2	29.4	31.8	29.8	25.6	49.5	44.1	47.3

feedback provide additional gains? **RQ4** (§4.5): Can solve-time routing effectively leverage specialized harness branches? **RQ5** (§4.6): Can human steering help under insufficient history signal?

4.2 Comparison with Baselines

Prior systems specialize on one metric cluster. A-Evolve leads the two pass-rate streams (45.2% CTF-Dojo, 47.5% FutureX) but covers only 21.1% of PolyBench markets. Meta-Harness leads all three PolyBench metrics (55.3% Coverage, 50.8% Accuracy, +320% Return) but falls below the no-evolution Sonnet baseline on FutureX (29.4% vs. 31.0%). Base solvers stay below 32.6% PolyBench Coverage on every model, and OctoTools, the frozen human-designed system, places third on PolyBench Return but does not lead any row. The pass-rate cluster and the portfolio cluster therefore sit in different baselines.

Our three variants jointly lead all five metrics. Without HITL intervention, the Full System combines multi-agent evolution with solve-time routing and reaches 97.9% PolyBench Coverage, 80.9% Accuracy, and 50.2% CTF-Dojo Pass. The Multi-agent variant leads FutureX at 49.5%, where evolving the right source and tooling matters more than per-task routing. The Adaptive variant leads PolyBench Return at +352%, where matching each market to a specialized strategy matters more than wider coverage.

4.3 Benchmark Bottlenecks

RQ2 asks whether Adaptive Auto-Harness targets the bottlenecks that limit each benchmark. We organize the analysis around the two losses introduced in §3.2: the *evolution loss* L_{evo} , the gap from capabilities the evolver class cannot construct; and the *adaptation loss* L_{adapt} , the gap from committing to a single harness across heterogeneous tasks.

Evolution bottlenecks (L_{evo}) differ across benchmarks. Figure 4 plots a benchmark-specific stress

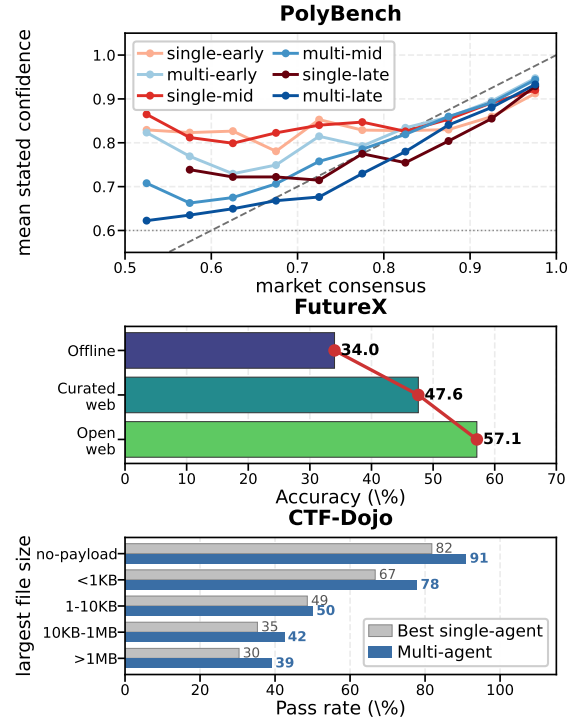


Figure 4: L_{evo} evidence across benchmark-specific bottlenecks. PolyBench stresses confidence calibration, FutureX stresses web-retrieval access, and CTF-Dojo stresses payload handling of different file sizes.

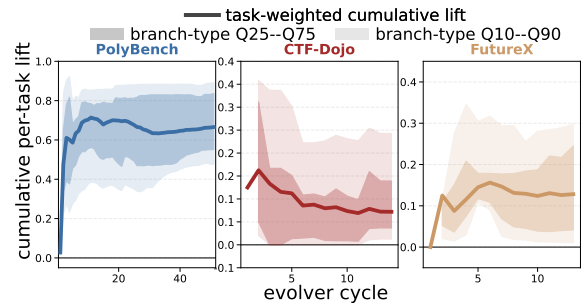


Figure 5: L_{adapt} evidence from branch specialization. Curves show cumulative routing lift over the baselines, while shaded bands show harness branch performance spread across cycles.

axis chosen to expose the most predictive capability. The PolyBench panel plots mean stated confidence against market consensus, defined as the implied

probability of the favored outcome from Polymarket prices; a well-calibrated harness tracks the diagonal. The multi-agent variant does so, while single-agent variants stay flat and over-state confidence on low-consensus markets, suggesting that the binding capability is consensus-aware confidence calibration rather than raw prediction skill. The FutureX panel plots pass rate against three retrieval tiers, ranging from offline to date-filtered Wikipedia plus DuckDuckGo to unrestricted DuckDuckGo. Pass rate increases monotonically from 34.0% to 47.6% to 57.1%, identifying source acquisition rather than reasoning as the binding capability. The CTF-Dojo panel plots pass rate against the largest challenge file size, binned into five tiers from no-payload to >1MB. The best single-agent variant declines from 81.8% to 30.4%, and the multi-agent variant declines from 90.9% to 39.1% while retaining roughly a 9-point margin throughout, indicating that payload-handling infrastructure becomes the binding capability as inputs grow and that the multi-agent evolver mitigates but does not eliminate this bottleneck. The binding capability therefore differs by benchmark, so no single fixed harness suffices.

Adaptation bottlenecks (L_{adapt}) remain after evolution. Figure 5 reports per-task routing lift over evolver cycles, where lift on each task is the routed-harness score minus the median score across single-committed-harness baselines. The solid curve is the within-cycle mean lift and the shaded band is performance variation across different branches; if a single committed harness were sufficient, the mean lift would approach zero as evolution progresses. We observe instead that the mean lift and variation remains positive across all cycles on all three benchmarks.

Takeaway. Although streams expose different bottlenecks, they can be addressed under the same principle: reduce evolution loss for missing harness capabilities and adaptation loss for task-specific fit.

4.4 Stateful Multi-Agent Evolution

RQ3 asks whether the four-role evolver in §3.3 (Analyst → parallel Researchers → Builder → Verifier) improves over a single-agent evolver, and whether its state channels matter. Figure 6 compares the full system with variants that remove temporal-reveal feedback or cross-cycle memory. The full system improves over the single-agent evolver on all three benchmarks: 20.3 → 44.3 CWR on PolyBench, 38% → 43% on CTF-Dojo, and 38% → 44% on FutureX. Removing memory causes the broad-

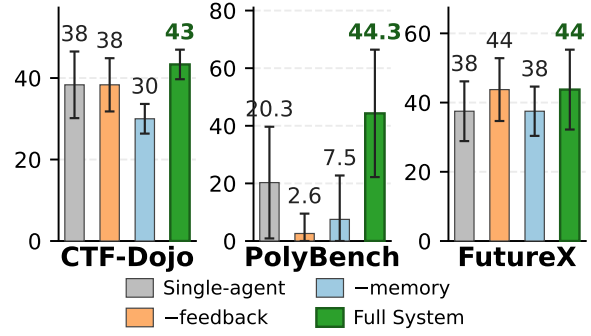


Figure 6: Ablations for multi-agent evolution. Removing evaluation feedback or cross-cycle memory degrades performance relative to the full stateful system.

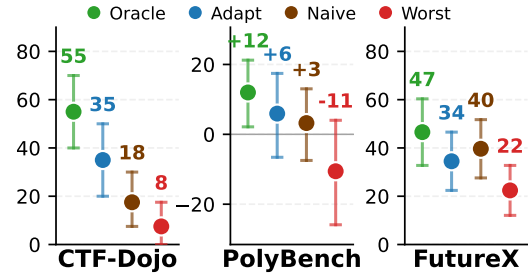


Figure 7: Solve-time routing evaluation on a evolved harness tree. Oracle reports the best branch performance per task, Adapt uses our agentic router, Naive always uses main workspace, and Worst reports the worst branch performance per task.

est degradation (2.6 CWR on PolyBench, 30% on CTF-Dojo, and 38% on FutureX), while removing feedback mainly hurts PolyBench (7.5 CWR), where outcomes resolve after trading. Overall, the four-role evolver is strongest when paired with persistent state: memory preserves cross-cycle search, and feedback turns resolved outcomes into later evolution signal.

4.5 Solve-Time Routing on the Harness Tree

RQ4 asks whether solve-time routing can leverage the specialized harness tree. We replay all branches on the same tasks and compare *Oracle* (best branch per task), *Adapt* (our router), *Naive* (main), and *Worst* (worst branch per task). Figure 7 shows clear Oracle-to-Naive headroom: 37 points on CTF-Dojo, +9 CWR on PolyBench, and 7 points on FutureX. Adapt captures the gap on CTF-Dojo and PolyBench, but trails Naive on FutureX, where source acquisition dominates routing choice.

Takeaway. Harness specialization creates headroom across streams, but FutureX shows that routing cannot compensate when source acquisition dominates branch choice.

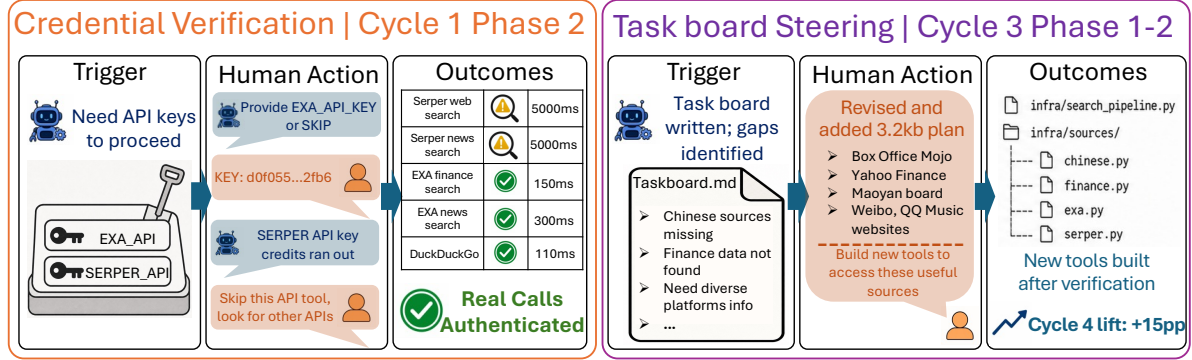
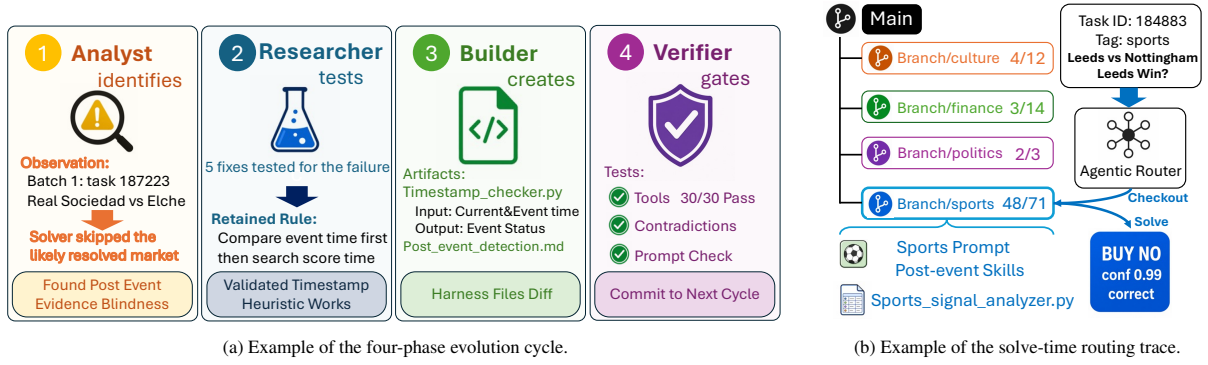


Figure 8: Extracted trajectories of the designed multi-agent evolution, agentic routing, and Human-in-the-Loop.

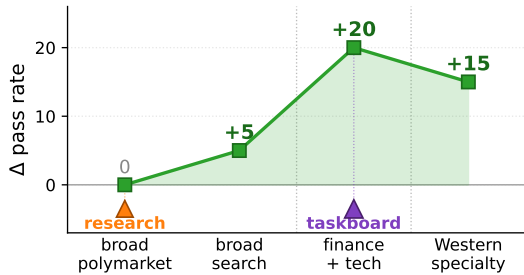


Figure 9: FutureX pass-rate lift over four task slices under two human-steering hooks. The orange triangle marks research-phase steering and the purple triangle marks task-board steering. Lift is measured against the no-HITL run.

4.6 Human Steering for Auto-Harnessing

RQ5 asks whether human steering helps when history lacks the source or access signal needed for evolution. Since all previous experiment does not apply HITL to ensure fairness, we evaluate this on FutureX and restrict human input to two hooks: *research-phase steering* supplies credentials when research is blocked, and *task-board steering* adds source directions the evolver cannot infer (Figure 8c). Figure 9 shows the slice-level effect: lift is 0 on broad polymarket questions, rises to +5 on broad search-dependent questions, peaks at +20 on the directly targeted finance&tech slice, and

remains +15 on adjacent Western-specialty questions. The pattern indicates that HITL helps when it injects the missing external signal, not generic human advice.

Takeaway. Human steering is most useful when the missing ingredient is external source knowledge rather than additional autonomous evolution.

5 Conclusions

Open-ended task streams expose three challenges for auto-harness deployment: unbounded task arrival, heterogeneous tasks, and non-stationarity. Adaptive Auto-Harness addresses these challenges by pairing sustained harness construction with solve-time task adaptation. The decomposition into evolution loss and adaptation loss clarifies why a single repeatedly updated harness is insufficient: the system must build missing capabilities from stream evidence while selecting the right specialized branch for each task. The experiments further show that the three mechanisms are complementary rather than interchangeable: multi-agent evolution constructs benchmark-specific capabilities, routing exploits harness specialization when branch signals are reliable, and human steering supplies external signals that history cannot contain.

6 Limitations

Benchmark coverage. We evaluate finite prefixes of three open-ended task streams: prediction markets, cybersecurity challenges, and event forecasting. These domains cover unbounded streams, task heterogeneity, and distributional non-stationarity, but the same framing should be tested further on additional deployment streams such as robotics, biomedical reasoning, and long-form writing.

Diagnostic losses. The evolution loss L_{evo} and adaptation loss L_{adapt} are analytical quantities, not directly estimated oracle losses. Our experiments diagnose them through bottleneck analyses, ablations, and routing controls rather than through a formal estimator of the oracle harness.

7 Ethics Statement

We use public or research benchmark tasks and do not introduce private user data. CTF-Dojo runs only inside isolated benchmark containers and does not target real systems. Human steering is limited to source guidance, task-board edits, and credential decisions; humans do not label answers or choose solver branches. Released artifacts should preserve sandbox restrictions and exclude credentials or provider secrets.

8 AI Usage Statement

We used AI assistants to refine the writing of this paper and to accelerate debugging and analysis during implementation and evaluation.

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A Benchmark and Evaluation Details

Across all three benchmarks, tasks are evaluated in chronological order. The solver receives task x_i with only the information available before its release time, while the evolver receives outcome labels only after the corresponding resolution time. The table generator loads the first record for each instance_id, so retries or duplicate logs do not change the reported metrics. The three datasets are finite evaluated prefixes of unbounded task streams

and jointly cover the three deployment challenges in §1: stream continuation (D1), within-stream heterogeneity (D2), and temporal shift (D3).

A.1 PolyBench

Source and composition. PolyBench is a Polymarket-derived prediction-market stream with 5,075 binary-outcome tasks from Feb 6–22, 2026. The stream spans politics, sports, finance, crypto, and entertainment markets, and each task is resolved against the official market outcome.

Temporal ordering. For each market, we store the task release timestamp and the outcome resolution timestamp. Retrieval evidence is filtered by release time during solving. Resolved labels are hidden from evolution until the corresponding market has resolved.

Metrics. Let $\mathcal{T}$ be the set of executed trades, excluding gated tasks, empty decisions, and SKIP decisions. Let s_i indicate whether trade i is correct, b_i be its confidence-weighted investment, and g_i be its realized profit. The metrics reported for PolyBench are:

$$\text{Acc} = 100 \cdot \frac{1}{N} \sum_{i \in \mathcal{T}} s_i$$

$$\text{CWR} = 100 \cdot \frac{\sum_{i \in \mathcal{T}} g_i}{\sum_{i \in \mathcal{T}} b_i}$$

$$\text{Return} = \frac{|\mathcal{T}|}{N} \cdot \text{CWR}.$$

Accuracy therefore rewards both broad coverage and correct decisions, while Return scales the portfolio profit-to-investment ratio by the fraction of tasks actually traded.

A.2 CTF-Dojo

Source and composition. CTF-Dojo is a 261-challenge security-competition stream drawn from pwncollege/ctf-archive, chronologically ordered from 2011 to 2024. Challenges cover binary exploitation, web security, cryptography, reverse engineering, and forensics, exposing changes in challenge style and tooling over time.

Sandbox and verification. Each challenge runs inside a per-task Docker sandbox with constrained network policy. Flags are submitted as text and verified by SHA-256 hash comparison against the official flag.

Metrics. We report Pass@1, the percentage of challenges solved within the benchmark budget. All CTF-Dojo aggregate results use the same chronological ordering as the released stream.

A.3 FutureX

Source and composition. FutureX is a 503-question event-forecasting stream over 82 days from Jan–Apr 2026, drawn from FutureX-Past. Questions cover finance, technology, geopolitics, and entertainment, with both English and Chinese-language variants. The zh-finance slice requires source discovery beyond default English-only retrieval.

Retrieval modes. *Strict* mode resolves all retrieval through the Wikipedia revision API restricted to revisions before the task release date. *Live* mode uses DuckDuckGo with a hard cutoff filter applied to result timestamps. The human-steering experiment in RQ5 uses live mode because authenticated providers such as Exa and Serper are gated behind credentials the evolver cannot synthesize from history.

Metrics. We report Pass@1 under the official FutureX criterion. Slice-level human-steering analyses use the same pass criterion and aggregate tasks by the groups shown in Figure 9.

B Implementation and Reproducibility Details

Execution protocol. All main runs use provider-hosted LLM APIs with native tool calling and the same chronological task order used by the benchmarks. Reported numbers are generated by evaluations/generate_main_table.py from the corresponding results.jsonl files. Closed provider-hosted models do not expose parameter counts; we therefore report the evaluated task counts and evolution cycles rather than GPU-hours. The full-system runs contain 5,075/261/503 solve trajectories and 51/14/26 evolution cycles for PolyBench/CTF-Dojo/FutureX, respectively.

Aggregation. Main-table values are single chronological runs rather than seed averages. This choice preserves the streaming protocol: random reshuffling would change the history available to the evolver and the router. The aggregation script is deterministic given the result files, and all percentages are computed over the full task stream unless otherwise stated.

Temporal reveal. Each task stores a release timestamp and, when available, a resolution timestamp. Solver calls are filtered against the release time. Evolution cycles receive the trajectory immediately but receive outcome feedback only after the task has resolved, so unresolved tasks remain unlabeled

history rather than leaked supervision.

Workspace artifacts. The evolver workspace persists a task board, research logs, verifier notes, tests, and architecture notes across cycles. These artifacts are separate from the solver workspace: the evolver may update harness files during evolution, while the solve-time router only inspects branch metadata and selects a branch for the incoming task.

Branch replay protocol. For routing analysis, every branch in the evolved harness tree is replayed on the same task set before constructing oracle and worst controls. Oracle and worst are therefore post-hoc diagnostic bounds; the deployed router only sees task context and branch metadata, not labels or branch outcomes.

Human-steering records. Human-steering events are author-provided system interventions rather than recruited human-subject annotations. Each event is logged with the triggering phase, requested external signal, and workspace location where the response is recorded. This makes steering auditable and keeps human input as source or access guidance rather than answer labels.

C Benchmark Non-Stationarity Diagnostics

Figures 10–12 provide benchmark-specific evidence that the temporal streams are not static IID pools. The diagnostics show how task properties change across time, explaining why a harness fitted to earlier batches can become mismatched to later tasks.

D Further Experiments

We include two supplemental diagnostics that clarify where the main gains come from without duplicating the RQ analyses in §4.

Evolver capability and construction budget. Figure 13 varies the evolver model and construction budget on CTF-Dojo. Stronger evolvers achieve higher pass rates, while additional budget mainly helps weaker evolvers and saturates for the strongest model.

Cross-domain workspace dilution. Figure 14 compares PolyBench CWR when using workspaces evolved on different domains. The PolyBench-specific workspace performs best, while the all-evolved workspace loses 57 points of CWR, showing that mixing heterogeneous experience can dilute domain-relevant harness structure.

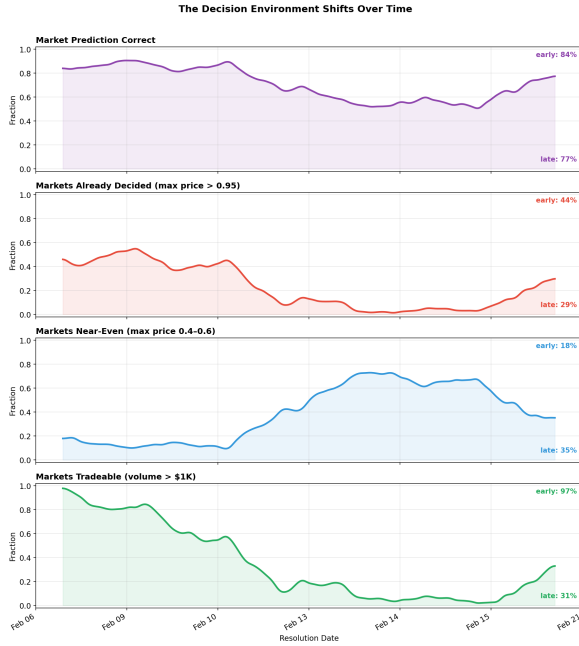


Figure 10: **PolyBench non-stationarity.** Strategy-relevant market properties vary across the temporal stream, so a harness fitted to earlier batches can become mismatched to later markets.

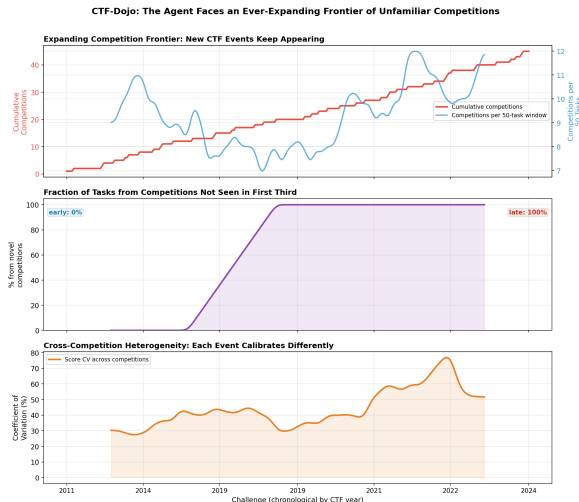


Figure 11: **CTF-Dojo non-stationarity.** Chronological ordering exposes changing challenge families and payload patterns rather than a fixed IID security-task pool.

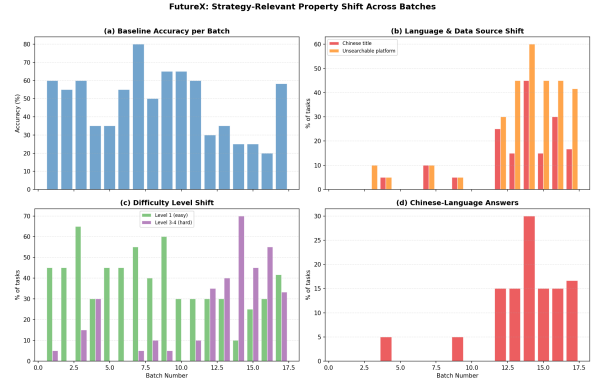


Figure 12: **FutureX non-stationarity.** Language, source, and difficulty properties shift across batches, creating solve-time harness mismatch when one static harness is reused for all tasks.

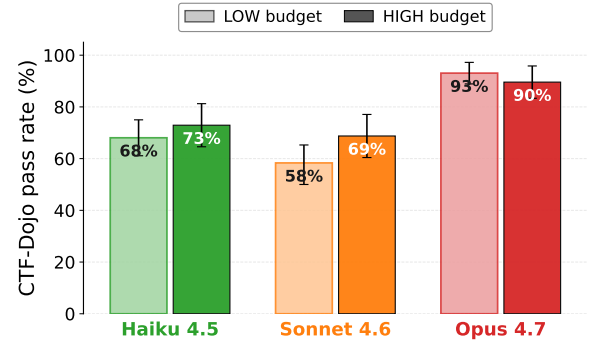


Figure 13: **Evolver capability on CTF-Dojo.** Pass rate improves with stronger evolver models; higher budget helps Haiku and Sonnet but gives little additional gain once Opus already reaches high performance.

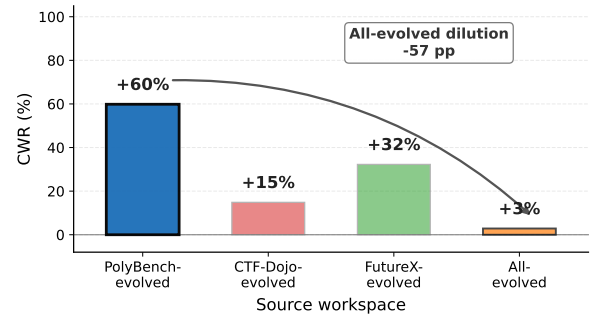


Figure 14: **PolyBench workspace dilution.** A PolyBench-evolved workspace reaches the highest CWR, while combining all evolved workspaces sharply reduces CWR, supporting the need for specialized harness branches rather than a single dense harness.